Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Dasilva Dairy Farms LLC #3

Physical address of dairy:

Number and Street	City	County	Zip Code
24197 E Mariposa RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 04/01/1993

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X207-X180-X005-XXXX

B. OPERATORS

DaSilva, Dennis

Operator name: DaSilva, Dennis

Telephone no.: (209) 000-0000 Landline, (209) 000-0000 Cellular

Mailing Address Number and Street	City	State	Zip Code
24666 Mariposa RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

DaSilva, Farms LLC

Operator name: DaSilva, Farms LLC

Telephone no.: (209) 000-0000 Landline, (209) 000-0000 Cellular

Mailing Address Number and Street	City	State	Zip Code
24666 Mariposa RD	Escalon	CA	95320

C. OWNERS

DaSilva, Farms LLC

Legal owner name: DaSilva, Farms LLC

Telephone no.: (209) 000-0000 Landline, (209) 000-0000 Cellular

Mailing Address Number and Street	City	State	Zip Code
24666 Mariposa RD	Escalon	CA	95320

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	425	0	0	0	0	0
Maximum number	425	0	0	0	0	0
Average number	425	0	0	0	0	0
Avg live weight (lbs)	1,300	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 76 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 11,202.98 *tons per reporting period*

Total nitrogen from manure: 146,449.42 *lbs per reporting period*

After ammonia losses (30% loss applied): 102,514.59 *lbs per reporting period*

Total phosphorus from manure: 24,850.05 *lbs per reporting period*

Total potassium from manure: 83,066.77 *lbs per reporting period*

Total salt from manure: 200,111.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 8,964,000 *gallons*

Total nitrogen generated: 21,938.47 *lbs*

Total phosphorus generated: 3,557.90 *lbs*

Total potassium generated: 18,719.17 *lbs*

Total salt generated: 138,433.54 *lbs*

	8,964,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	8,964,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well	Ground water
Domestic Well/ House	Ground water
Domestic Well/ Milk House	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
F1	39	39	2	both	X207-X180-X005-XXXX
F2	28	28	2	both	X207-X190-X044-XXXX
F3	24	24	0	none	X207-X180-X002-XXXX
F4	97	50	0	none	X207-X180-X002-XXXX
F5	9	9	0	none	X207-X190-X019-XXXX
F6 (Dusi)	40	40	2	both	X207-X180-X004-XXXX
F7	15	15	0	none	X207-X350-X006-XXXX
Totals for areas that were used for application	107	107	6		
Totals for areas that were not used for application	145	98	0		
Land application area totals	252	205	6		

B. CROPS AND HARVESTS

F1

Field name: F1

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	702.00 *ton*	Dry-weight		79.7	20,800.00	2,100.00	25,900.00		20.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	152.01	15.35	189.28	1,461.60

F1

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	1,170.00 *ton*	Dry-weight		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	0.00	0.00	0.00	0.00

F2

Field name: F2

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 28 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	504.00 *ton*	Dry-weight		76.1	37,300.00	2,880.00	34,600.00		20.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	320.93	24.78	297.70	1,720.80

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	840.00 *ton*	Dry-weight		72.0	13,100.00	2,740.00	14,200.00		1.56

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	220.08	46.03	238.56	262.08

F6 (Dusi)

Field name: F6 (Dusi)

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	720.00 *ton*	Dry-weight		75.7	18,600.00	1,370.00	17,200.00		20.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	162.71	11.98	150.47	1,758.35

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	1,200.00 *ton*	Dry-weight		67.0	14,400.00	1,870.00	8,270.00		5.76

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	285.12	37.03	163.75	1,140.48

NUTRIENT BUDGET

A. LAND APPLICATIONS

F1 - 10/30/2022: Oats, silage-soft dough

Field name: F1

Crop: Oats, silage-soft dough Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	84.76	275.14	54.25	3,325.20	156.00 *ton*
Application event totals		84.76	275.14	54.25	3,325.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	69.79	6.44	43.08	415.97	432,000.00 *gal*
Ag Well	Ground water	9.12	0.00	0.00	177.48	1,296,000.00 *gal*
Application event totals		78.91	6.44	43.08	593.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.90	2.22	21.91	129.41	216,000.00 *gal*
Ag Well	Ground water	9.12	0.00	0.00	177.48	1,296,000.00 *gal*
Application event totals		36.02	2.22	21.91	306.89	

F1 - 06/02/2023: Corn, silage

Field name: F1

Crop: Corn, silage Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

F1 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.69	8.98	36.61	249.58	972,000.00 *gal*
Ag Well	Ground water	36.49	0.00	0.00	709.92	5,184,000.00 *gal*
Application event totals		70.19	8.98	36.61	959.50	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	30.41	0.00	0.00	591.60	4,320,000.00 *gal*
Application event totals		30.41	0.00	0.00	591.60	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.95	7.99	32.54	221.85	864,000.00 *gal*
Ag Well	Ground water	27.37	0.00	0.00	532.44	3,888,000.00 *gal*
Application event totals		57.32	7.99	32.54	754.29	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.46	5.99	24.40	166.39	648,000.00 *gal*
Ag Well	Ground water	27.37	0.00	0.00	532.44	3,888,000.00 *gal*
Application event totals		49.83	5.99	24.40	698.82	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	24.33	0.00	0.00	473.28	3,456,000.00 *gal*
Application event totals		24.33	0.00	0.00	473.28	

F1 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	24.33	0.00	0.00	473.28	3,456,000.00 *gal*
Application event totals		24.33	0.00	0.00	473.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	18.25	0.00	0.00	354.96	2,592,000.00 *gal*
Application event totals		18.25	0.00	0.00	354.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	18.25	0.00	0.00	354.96	2,592,000.00 *gal*
Application event totals		18.25	0.00	0.00	354.96	

F2 - 10/30/2022: Oats, silage-soft dough

Field name: F2

Crop: Oats, silage-soft dough

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	211.90	687.86	135.62	8,313.00	280.00 *ton*
Application event totals		211.90	687.86	135.62	8,313.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.21	8.97	60.00	579.38	432,000.00 *gal*
Ag Well	Ground water	10.59	0.00	0.00	206.00	1,080,000.00 *gal*
Application event totals		107.80	8.97	60.00	785.38	

F2 - 10/30/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.93	6.19	61.03	360.50	432,000.00 *gal*
Ag Well	Ground water	10.59	0.00	0.00	206.00	1,080,000.00 *gal*
Application event totals		85.52	6.19	61.03	566.51	

F2 - 06/02/2023: Corn, silage

Field name: F2

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	31.29	8.34	33.99	231.75	648,000.00 *gal*
Ag Well	Ground water	33.89	0.00	0.00	659.21	3,456,000.00 *gal*
Application event totals		65.17	8.34	33.99	890.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	25.42	0.00	0.00	494.41	2,592,000.00 *gal*
Application event totals		25.42	0.00	0.00	494.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.86	5.56	22.66	154.50	432,000.00 *gal*
Ag Well	Ground water	25.42	0.00	0.00	494.41	2,592,000.00 *gal*
Application event totals		46.27	5.56	22.66	648.91	

F2 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.86	5.56	22.66	154.50	432,000.00 *gal*
Ag Well	Ground water	25.42	0.00	0.00	494.41	2,592,000.00 *gal*
Application event totals		46.27	5.56	22.66	648.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.86	5.56	22.66	154.50	432,000.00 *gal*
Ag Well	Ground water	25.42	0.00	0.00	494.41	2,592,000.00 *gal*
Application event totals		46.27	5.56	22.66	648.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	21.18	0.00	0.00	412.00	2,160,000.00 *gal*
Application event totals		21.18	0.00	0.00	412.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	21.18	0.00	0.00	412.00	2,160,000.00 *gal*
Application event totals		21.18	0.00	0.00	412.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	21.18	0.00	0.00	412.00	2,160,000.00 *gal*
Application event totals		21.18	0.00	0.00	412.00	

F6 (Dusi) - 11/02/2022: Oats, silage-soft dough

Field name: F6 (Dusi)

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

F6 (Dusi) - 11/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	84.76	275.14	54.25	3,325.20	160.00 *ton*
Application event totals		84.76	275.14	54.25	3,325.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.05	6.28	42.00	405.57	432,000.00 *gal*
Ag Well	Ground water	8.90	0.00	0.00	173.04	1,296,000.00 *gal*
Application event totals		76.94	6.28	42.00	578.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	39.34	3.25	32.04	189.26	324,000.00 *gal*
Ag Well	Ground water	8.90	0.00	0.00	173.04	1,296,000.00 *gal*
Application event totals		48.24	3.25	32.04	362.31	

F6 (Dusi) - 06/04/2023: Corn, silage

Field name: F6 (Dusi)

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	105.95	343.93	67.81	4,156.50	200.00 *ton*
Application event totals		105.95	343.93	67.81	4,156.50	

F6 (Dusi) - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	32.85	8.76	35.69	243.34	972,000.00 *gal*
Ag Well	Ground water	35.58	0.00	0.00	692.17	5,184,000.00 *gal*
Application event totals		68.43	8.76	35.69	935.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	29.65	0.00	0.00	576.81	4,320,000.00 *gal*
Application event totals		29.65	0.00	0.00	576.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.20	7.79	31.72	216.30	864,000.00 *gal*
Ag Well	Ground water	26.69	0.00	0.00	519.13	3,888,000.00 *gal*
Application event totals		55.89	7.79	31.72	735.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	14.60	3.89	15.86	108.15	432,000.00 *gal*
Ag Well	Ground water	26.69	0.00	0.00	519.13	3,888,000.00 *gal*
Application event totals		41.29	3.89	15.86	627.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	23.72	0.00	0.00	461.45	3,456,000.00 *gal*
Application event totals		23.72	0.00	0.00	461.45	

F6 (Dusi) - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	23.72	0.00	0.00	461.45	3,456,000.00 *gal*
Application event totals		23.72	0.00	0.00	461.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	17.79	0.00	0.00	346.08	2,592,000.00 *gal*
Application event totals		17.79	0.00	0.00	346.08	

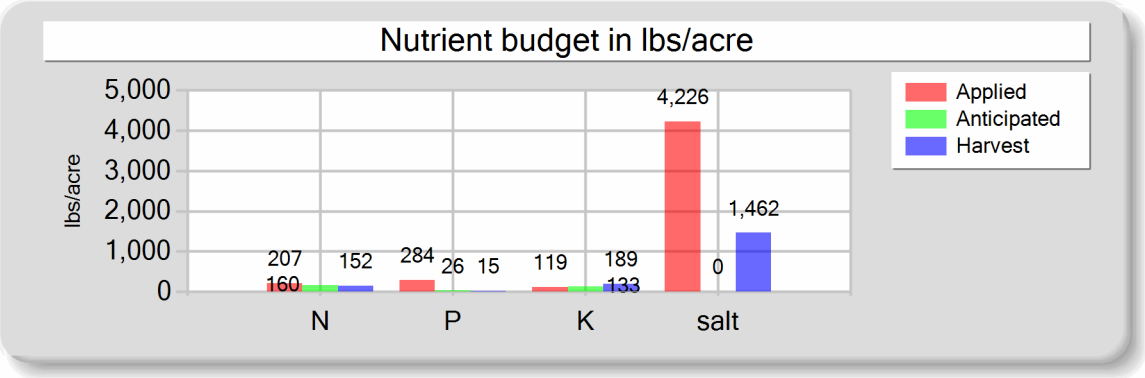
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	17.79	0.00	0.00	346.08	2,592,000.00 *gal*
Application event totals		17.79	0.00	0.00	346.08	

B. NUTRIENT BUDGET

F1 - 10/30/2022: Oats, silage-soft dough

Field name: F1 Crop: Oats, silage-soft dough Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	84.76	275.14	54.25	3,325.20
Process wastewater	96.69	8.67	64.98	545.38
Fresh water	18.25	0.00	0.00	354.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.70	283.81	119.23	4,225.54
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	152.01	15.35	189.28	1,461.60
Nutrient balance	54.69	268.46	-70.05	2,763.94
Applied to removed ratio	1.36	18.49	0.63	2.89

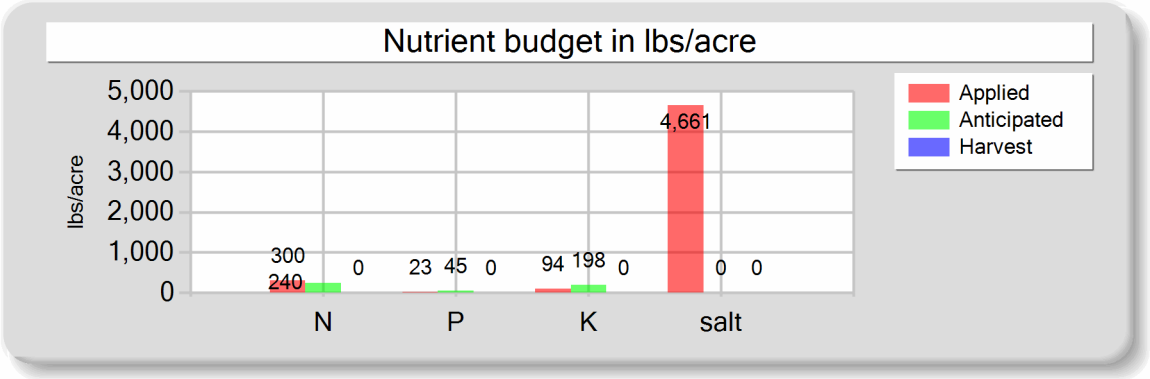
Fresh water applied	
2,592,000.00	*gallons*
95.45	*acre-inches*
2.45	*inches/acre*

Process wastewater applied	
648,000.00	*gallons*
23.86	*acre-inches*
0.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

F1 - 06/02/2023: Corn, silage

Field name: F1 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	86.10	22.96	93.55	637.81
Fresh water	206.80	0.00	0.00	4,022.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	299.90	22.96	93.55	4,660.67
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	299.90	22.96	93.55	4,660.67
Applied to removed ratio	0.00	0.00	0.00	0.00

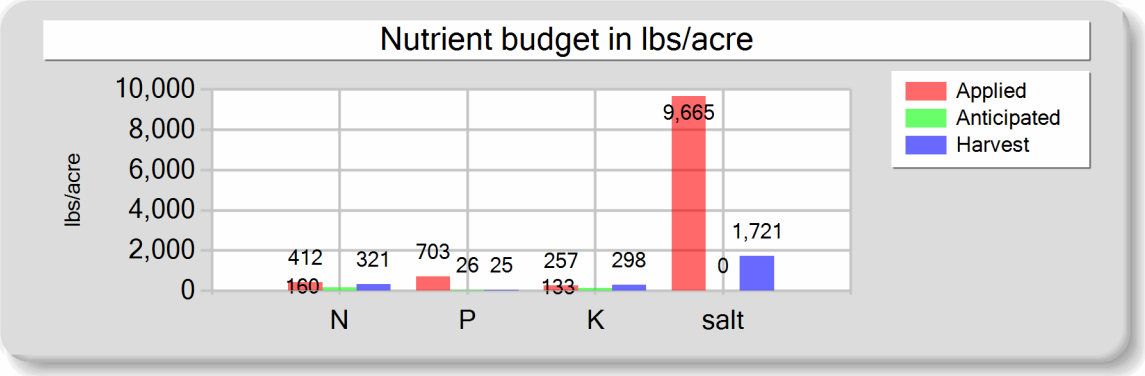
Fresh water applied	
29,376,000.00	*gallons*
1,081.82	*acre-inches*
27.74	*inches/acre*

Process wastewater applied	
2,484,000.00	*gallons*
91.48	*acre-inches*
2.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

F2 - 10/30/2022: Oats, silage-soft dough

Field name: F2 Crop: Oats, silage-soft dough Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	211.90	687.86	135.62	8,313.00
Process wastewater	172.14	15.17	121.03	939.89
Fresh water	21.18	0.00	0.00	412.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	412.22	703.03	256.64	9,664.89
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	320.93	24.78	297.70	1,720.80
Nutrient balance	91.29	678.25	-41.06	7,944.09
Applied to removed ratio	1.28	28.37	0.86	5.62

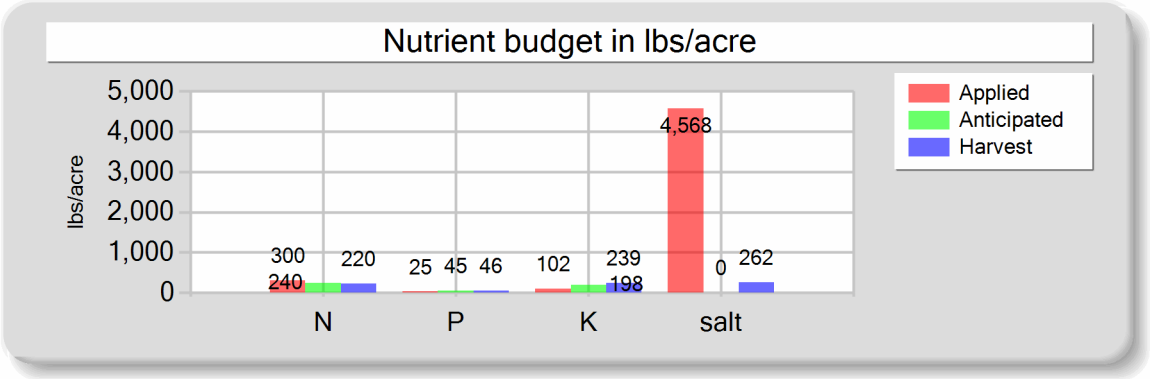
Fresh water applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
2.84	*inches/acre*

Process wastewater applied	
864,000.00	*gallons*
31.82	*acre-inches*
1.14	*inches/acre*

Total harvests for the crop	
1	*harvests*

F2 - 06/02/2023: Corn, silage

Field name: F2 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	93.86	25.03	101.97	695.26
Fresh water	199.09	0.00	0.00	3,872.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	299.95	25.03	101.97	4,568.10
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	220.08	46.03	238.56	262.08
Nutrient balance	79.87	-21.00	-136.59	4,306.02
Applied to removed ratio	1.36	0.54	0.43	17.43

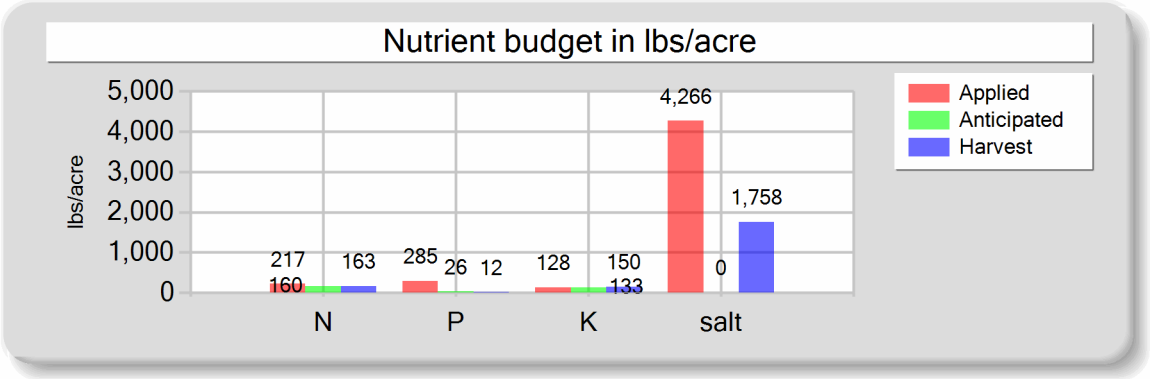
Fresh water applied	
20,304,000.00	*gallons*
747.73	*acre-inches*
26.70	*inches/acre*

Process wastewater applied	
1,944,000.00	*gallons*
71.59	*acre-inches*
2.56	*inches/acre*

Total harvests for the crop	
1	*harvests*

F6 (Dusi) - 11/02/2022: Oats, silage-soft dough

Field name: F6 (Dusi) Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	84.76	275.14	54.25	3,325.20
Process wastewater	107.39	9.53	74.04	594.83
Fresh water	17.79	0.00	0.00	346.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	216.94	284.68	128.28	4,266.12
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	162.71	11.98	150.47	1,758.35
Nutrient balance	54.22	272.69	-22.18	2,507.77
Applied to removed ratio	1.33	23.75	0.85	2.43

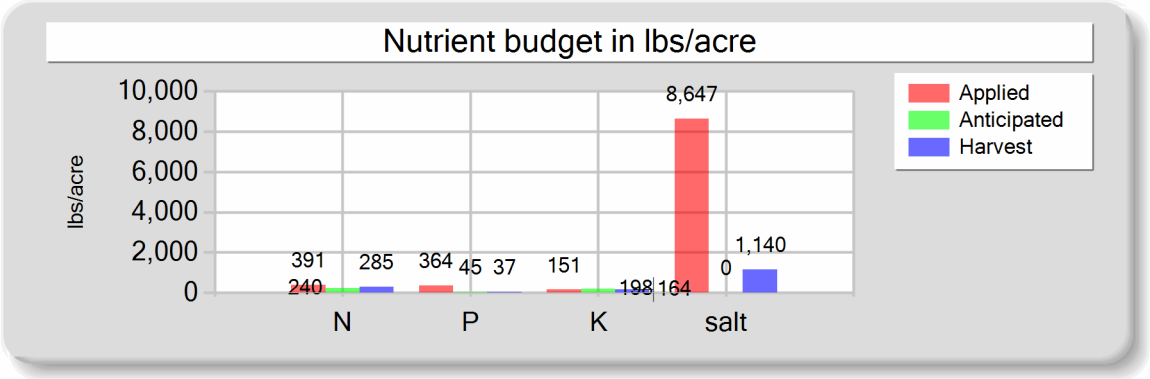
Fresh water applied	
2,592,000.00	*gallons*
95.45	*acre-inches*
2.39	*inches/acre*

Process wastewater applied	
756,000.00	*gallons*
27.84	*acre-inches*
0.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

F6 (Dusi) - 06/04/2023: Corn, silage

Field name: F6 (Dusi) Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	105.95	343.93	67.81	4,156.50
Process wastewater	76.65	20.44	83.28	567.79
Fresh water	201.63	0.00	0.00	3,922.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	391.23	364.37	151.08	8,646.58
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	285.12	37.03	163.75	1,140.48
Nutrient balance	106.11	327.34	-12.66	7,506.10
Applied to removed ratio	1.37	9.84	0.92	7.58

Fresh water applied	
29,376,000.00	*gallons*
1,081.82	*acre-inches*
27.05	*inches/acre*

Process wastewater applied	
2,268,000.00	*gallons*
83.52	*acre-inches*
2.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure

Sample and source description: Manure

Sample date: 10/25/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 18.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	42,200.00	8,320.00	110,000.00	8,900.00	2,700.00	231.00	519.00		51.00
DL	0.50	0.50	0.50	5.00	5.00	5.00	1.00	0.50		0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2302045-01

Sample and source description: Lagoon Water 2302045-01

Sample date: 02/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	755.00	755.00			69.70	466.00								4,850.00	4,500
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2303409-01

Sample and source description: Lagoon Water 2303409-01

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	582.00	582.00			48.10	474.00								4,860.00	2,800
DL	0.60	10.00			0.10	10.00								1.00	1

Lagoon Water 2307243-01

Sample and source description: Lagoon Water 2307243-01

Sample date: 07/17/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	162.00	159.00			43.20	176.00								2,310.00	1,200
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2310043-01

Sample and source description: Lagoon Water 2310043-01

Sample date: 09/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	452.00	352.00			103.00	441.00								4,570.00	2,500
DL	0.60	0.60			0.10	10.00								1.00	1

C. FRESH WATER ANALYSES

Ag Well

2401171-03

Sample description: 2401171-03

Sample date: 01/12/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	32.90										1,160.00	640
DL	1.40										1.00	1

Domestic Well/ House

Domestic Well/ House

2401171-01

Sample description: 2401171-01

Sample date: 01/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			12.70								1,130.00	
DL			0.40								1.00	

Domestic Well/ Milk House

2401171-02

Sample description: 2401171-02

Sample date: 01/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			7.28								1,160.00	
DL			0.40								1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

F1 - 10/30/2022: Oats, silage-soft dough

F1 - 10/30/2022: Oats, silage-soft dough

2305234-01

Sample and source description: 2305234-01

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,800.00	2,100.00	25,900.00		20.00
DL	0.50	0.50	0.50		0.10

F2 - 10/30/2022: Oats, silage-soft dough

2305234-02

Sample and source description: 2305234-02

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	37,300.00	2,880.00	34,600.00		20.00
DL	0.50	0.50	0.50		0.10

F2 - 06/02/2023: Corn, silage

2310282-02

Sample and source description: 2310282-02

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	2,740.00	14,200.00		1.56
DL	0.50	0.50	0.50		0.10

F6 (Dusi) - 11/02/2022: Oats, silage-soft dough

2305234-03

Sample and source description: 2305234-03

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,600.00	1,370.00	17,200.00		20.10
DL	0.50	0.50	0.50		0.10

F6 (Dusi) - 06/04/2023: Corn, silage

2310037-01

Sample and source description: 2310037-01

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	1,870.00	8,270.00		5.76
DL	0.50	0.50	0.50		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

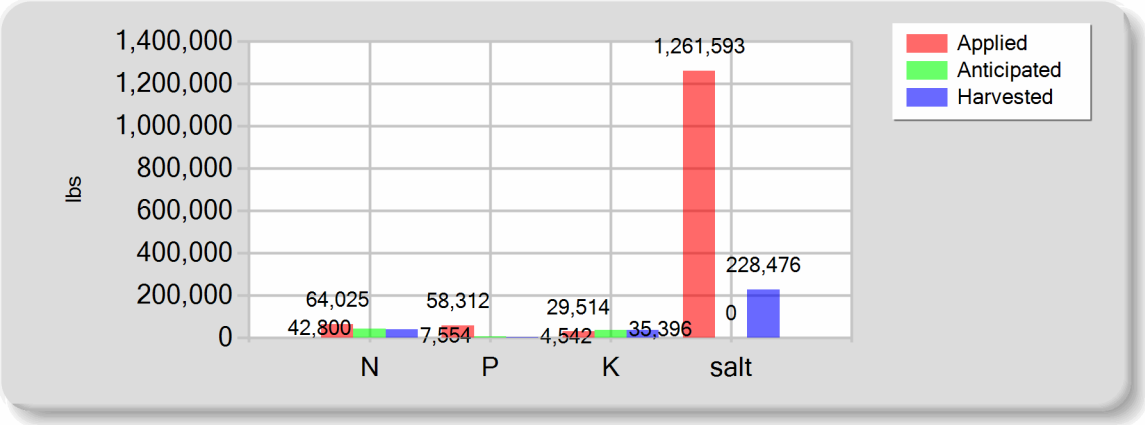
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

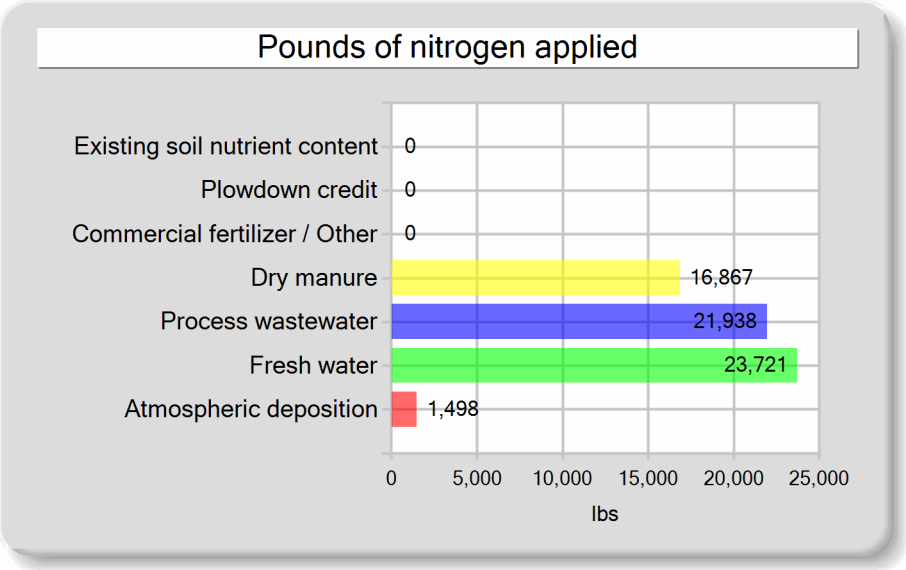
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

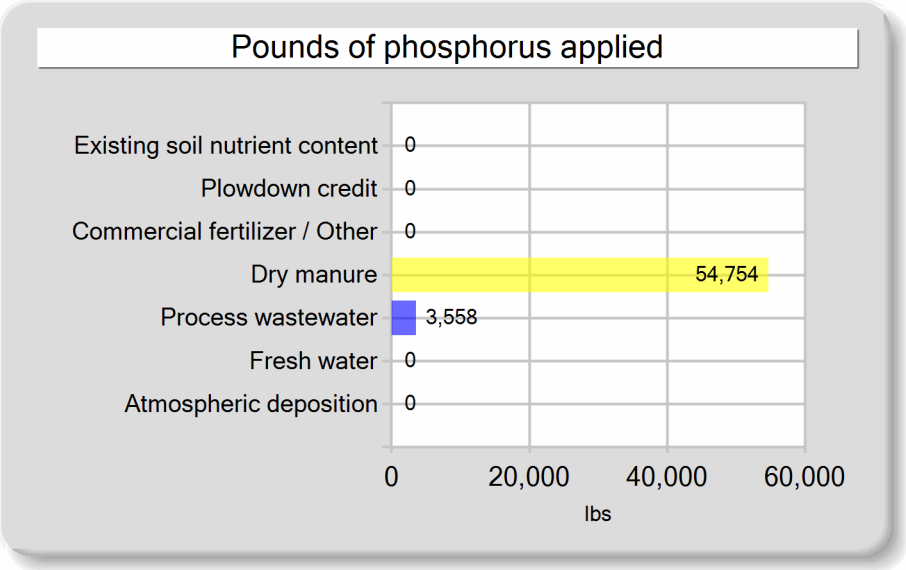
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	16,867.24	54,753.66	10,795.03	661,714.80
Process wastewater	21,938.47	3,557.90	18,719.17	138,433.54
Fresh water	23,721.16	0.00	0.00	461,445.12
Atmospheric deposition	1,498.00	0.00	0.00	0.00
Total nutrients applied	64,024.87	58,311.56	29,514.20	1,261,593.46
Anticipated crop nutrient removal	42,800.00	7,554.20	35,395.60	0.00
Actual crop nutrient removal	38,989.82	4,541.68	34,965.51	228,476.16
Nutrient balance	25,035.05	53,769.88	-5,451.31	1,033,117.30
Applied to removed ratio	1.64	12.84	0.84	5.52

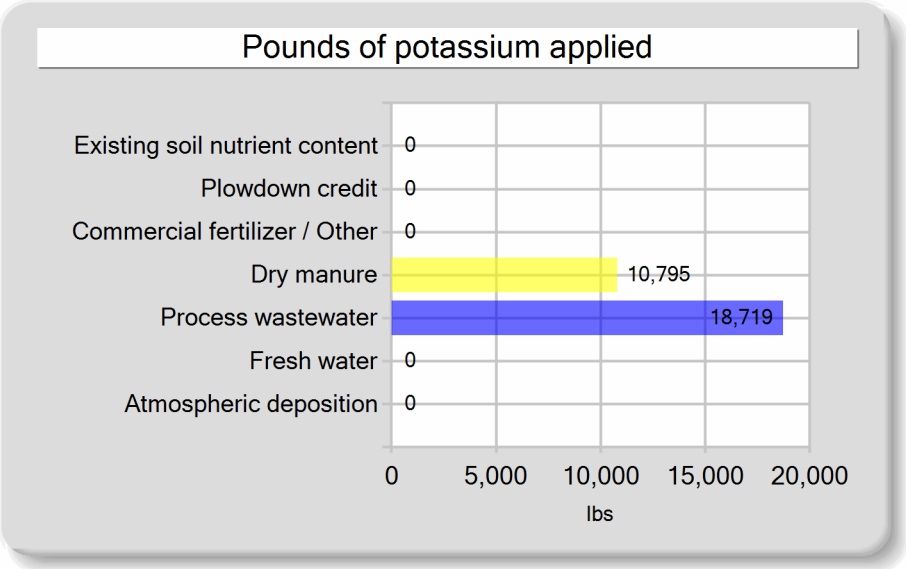
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

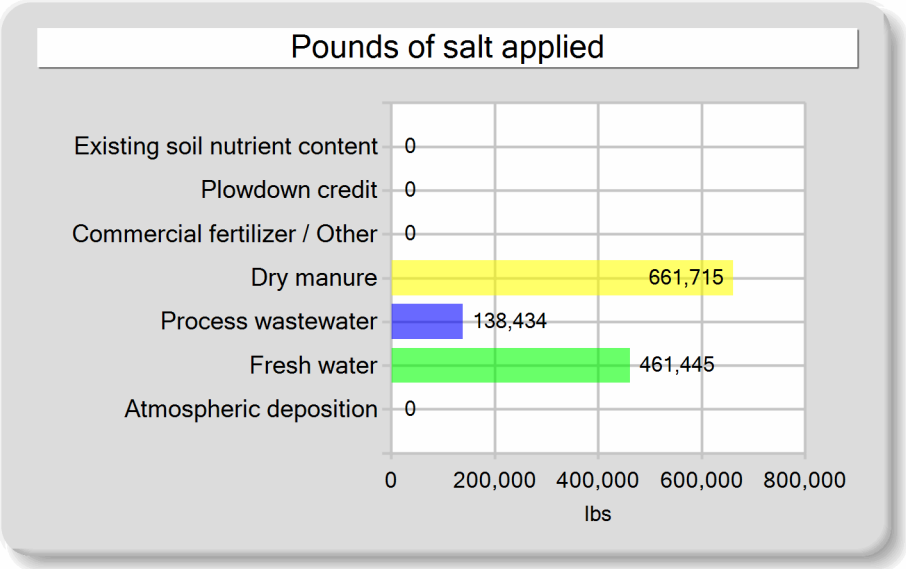


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

*Field #1 Corn Silage was not sample in 2023, we will be sure to sample in 2024.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

X	X
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Farms LLC DaSilva	Dennis DaSilva
PRINT OR TYPE NAME	PRINT OR TYPE NAME
✓ 07/02/2024	✓ 07/02/2024
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	N/A
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	N/A
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	